FOLLOW-ON DEMO

What changes when CoWork adds semantic views and RelationalAI?

Same telco recovery question. Three paths tested: CoWork, CoWork with source-only semantic views, and CoWork with semantic views plus a RelationalAI-backed Cortex agent.

1

CoWork

Useful analysis plan from natural language.

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2

CoWork + semantic views

Governed source-only metrics.

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3

CoWork + semantic views + RelationalAI

Exact computed answer with reasoner provenance.

THE TEST

Use one decision-heavy business question

Which towers should we upgrade, why those towers, and what constraint limits the plan?

Not just SQL

It needs a decision

The user is not asking for a table scan. They want an upgrade plan under budget and install-week constraints.

Not just text

It needs computed outputs

Risk, impact, call paths, upgrade tiers, and objective tradeoffs have to exist somewhere.

Not just a demo

It needs provenance

Customers need to inspect the answer and change the logic when business policy changes.

PATH 1: COWORK

Actual answer: a 24-tower heuristic plan from raw tables

CoWork inspected the raw schema, built a composite 0-100 urgency score, and selected 24 towers under the $5M capex envelope.

It identifies fixed capex as binding: $4,947,688 spent, about $52K remaining, and tower #25 would push the plan over budget.

Raw-table heuristic

What it used

- Tower health from equipment health scores.
- Advisory severity from recalls, defects, EOL, and firmware bugs.
- Subscriber impact from unique subscribers and LTV at risk.
- Call graph quality and dropped-call exposure.
- Churn exposure from affected subscribers.

PATH 2: COWORK + SEMANTIC VIEWS

Actual answer: a source-only heuristic plan, but not a proven optimum

55
towers recommended

$4.93M
planned spend

541
Gbps claimed capacity added

capex
binding constraint

What Snowsight produced from TELCO_SOURCE_NAIVE_SV

Claim	Value
Selected towers	55 of 250
Total spend	$4,929,904
Budget remaining	$70,096
Priority method	Composite 0-1 score across five source signals
Limiting constraint	Fixed capex budget

Why those towers

Segment	Explanation
Top 14	All DEGRADED, all WEST, with high latency, packet loss, traffic, churn-risk interactions, and recall exposure.
Towers 15-55	ACTIVE or MAINTENANCE towers with heavy advisory exposure and moderate call volume.
Deferred 195	Lower average composite score; generally healthier, fewer advisories, less urgent.

Actual Snowsight semantic answer; source-only, no RelationalAI result tables

THE KEY DIFFERENCE

Semantic views describe known business measures. RelationalAI adds executable business logic.

Semantic view

What it does well

Defines business names, dimensions, facts, metrics, and verified SQL patterns so a Snowflake agent can ask cleaner analytical questions.

Boundary

What it does not do

It can rank observed metrics, but it does not train a failure model, run graph impact, enumerate paths, or solve the constrained upgrade plan.

RelationalAI

What it adds

It creates those outputs from a durable model: rules, relationships, graph algorithms, prediction, path reasoning, and optimization.

In the source-only test, the semantic view gives a credible budget-constrained heuristic plan. It still does not prove optimality against predicted failure risk, graph-derived customer impact, and tier tradeoffs.

Actual answer: the computed recovery plan

OPTIMAL	$4.99M	183	budget
solver status	total cost	Gbps restored	binding constraint

RAI_QUERY_MODEL

plan_summary

Plan ID	TELCO_RECOVERY_STANDALONE
Towers covered	27
Install weeks	167
Tier mix	Gold 8, Silver 8, Bronze 11

Why this matters

The answer is no longer a plan to build analysis

It is a deployed reasoning result. The agent tool returns the optimized plan and can point to the catalog query and physical result tables behind it.

REASONER COVERAGE

The RAI answer carries the reasoner chain with it

Prediction

Equipment failure

Top predicted risk: EQP-00361 at 99.230% failure probability.

Graph

Customer impact

Tower impact includes active subscribers, weighted impact, and PageRank-weighted influence.

Path

Influential route

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SUB-CON-00900 -> SUB-CON-00814 ->
SUB-ENT-0038 -> SUB-CON-00644
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Optimization

Upgrade plan

27 selected towers, $4,985,789 spend, 183 Gbps restored.

Rules

Business logic

Rules and curated tools define what counts as risk, impact, and recovery priority.

Provenance

Inspectable outputs

Catalog tools read named tables: RESTORE_PLAN, SELECTED_UPGRADES, TOWER_FAILURE_SCORE, TOWER_CUSTOMER_IMPACT, TOP_CALL_PATH.

Same Snowflake UX, deeper reasoning behind the answer

SIDE-BY-SIDE

Actual difference in answer quality

Need	CoWork	CoWork + semantic views	CoWork + semantic views + RelationalAI
Answer type	Raw-table heuristic plan.	Source-only semantic heuristic plan.	Computed decision result from catalog tools.
Exact plan	Returns 24 towers for $4.95M.	Returns a 55-tower source-metric heuristic plan for $4.93M.	Creates and returns the optimal plan: $4.99M, 27 towers, 183 Gbps.
Why those towers	Composite urgency score from health, advisories, subscriber impact, call quality, and churn.	Composite score across health, advisories, performance, subscriber impact, and cost efficiency.	Combines prediction, graph impact, path reasoning, rules, and optimization.
Constraint	Identifies fixed capex as limiting, with ~$52K remaining.	Identifies fixed capex as limiting, with ~$70K remaining.	Solves under constraints and returns binding constraint: budget.
Correction path	Change prompt, SQL, or generated code.	Change dimensions, metrics, descriptions, or verified queries.	Change model logic, relationships, objective, constraints, or curated tools, then redeploy.

RAI AS EVALUATOR

RelationalAI can score candidate plans, not just produce its own.

Plan	Status	Towers	Cost	Weeks	Capacity	Tier mix	RAI scoring
Actual Snowsight CoWork answer	Credible heuristic; row-level scoring pending	24	$4.95M	Not reported	Not reported	Implied Silver-tier pricing	Needs selected tower/tier rows
Actual Snowsight semantic answer	Credible heuristic; row-level scoring pending	55	$4.93M	Not reported	541 Gbps claimed	Not captured	Needs selected tower/tier rows
CoWork + semantic views + RelationalAI	Optimal	27	$4.99M	167	183 Gbps	8 Gold, 8 Silver, 11 Bronze	100% benchmark

RAI evaluates candidate plans against the governed objective: restored capacity multiplied by predicted failure risk and graph-derived customer impact, under the $5M budget and 200 install-week envelope.

WHY THE HEURISTICS LOSE

The semantic answer is useful, but RAI can still explain what it does not prove.

Suboptimality

What RAI explains

The CoWork and semantic answers rank source-observable signals and identify the capex constraint. To score either one exactly, RAI needs the returned tower/tier rows as candidate plans.

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BERTOLANI.TELCO_PLAN_EVALUATION.PLAN_EXPLANATIONS
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Question	Semantic answer	RAI-added check
Is it budget aware?	Yes. CoWork finds $4.95M; semantic finds $4.93M. Both identify capex as binding.	Verify hard constraints against row-level selections.
Is it optimal?	Not proven. It uses a composite heuristic score.	Compare to the prescriptive optimum and quantify the objective gap.
Is it explainable?	Yes for source metrics and advisories.	Add prediction, graph impact, paths, tier tradeoffs, and provenance.

This is the added benefit: Snowflake can generate and govern candidate analysis; RelationalAI can judge it, explain the gap, and persist the corrected business logic.

CLOSING FRAME

Be explicit: Snowflake is the platform, RelationalAI adds decision reasoning

Snowflake

Governed AI data platform

Data, permissions, Cortex, agents, semantic views, and the Snowsight experience.

RelationalAI

Reasoning layer

Ontology, rules, graph, paths, prediction, optimization, provenance, and durable correction.

Customer gain

More complete answers

Not just "how would I analyze this?" but "here is the computed decision, why it won, and where to change it."

"We are not replacing the Snowflake agent experience. We are extending what that agent can accurately answer when the question requires connected business reasoning."